

MECE - 101

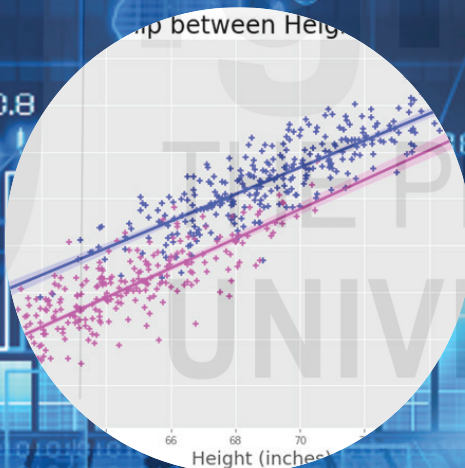
INTRODUCTORY ECONOMETRIC METHODS

$$E(u_i) = 0$$

$$E(u_i^2) = \sigma^2$$

$$E(u_i u_j) = 0; \quad i \neq j$$

$$E(X_i u_i) = 0$$



INTRODUCTORY ECONOMETRIC METHODS



**School of Social Sciences
Indira Gandhi National Open University
Maidan Garhi, New Delhi-110068**

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Unit 2	Review of Statistical Foundations of Econometrics	Dr. Sudip Mukherjee, Dinabandhu Mahavidyalaya, Bongaon, West Bengal
Unit 2	Review of Matrix Algebra	Mr. Debasish Dash, Independent Researcher, Delhi
Block 2	Classical Regression Model	
Unit 4	Estimation of Two-variable Regression Model	Dr. Samraj Sahay, Independent Researcher, Delhi
Unit 5	Residual Analysis	Dr. Gurleen Kaur, Assistant Professor of Economics, Guru Gobind Singh College of Commerce, University of Delhi
Unit 6	Estimation of Multiple Regression Models	Prof. Kaustuva Barik, IGNOU, New Delhi
Unit 7	Evaluation of Multiple Regression Models	Dr. Samraj Sahay, Independent Researcher, Delhi
Block 3	Violations of Basic Assumptions	
Unit 8	Model Specification Issues	Dr. Sahba Fatima, Independent Researcher, Lucknow
Unit 9	Autocorrelation	Dr. Priti Agarwal, Jindal School of Government and Public Policy, OP Jindal Global University, Sonapat, Haryana (Units 9, 10 and 11)
Unit 10	Multicollinearity	
Unit 11	Heteroscedasticity	
Unit 12	Errors in Variables	Prof. Shibalal Meher (retd.), Nabakrushna Choudhury Centre for Development Studies, Bhubaneswar
Unit 13	Stochastic Regressors	Dr. Narendra Nath Dalei, Associate Professor, Central University of Himachal Pradesh, Dharamshala
Block 4	Extensions of Regression Models	
Unit 14	Qualitative Independent Variables in OLS Models	Dr. Manish Gupta, Associate Professor, NIPFP, New Delhi
Unit 15	Qualitative Dependent Variables in OLS Models	Dr. Akshaya Kumar Mohanty, Associate Professor, Mizoram University, Aizawl
Unit 16	Introduction to Simultaneous Equations Models	Prof. Biswajit Nag and Ms. Satavisha Mukherjee, Indian Institute of Foreign Trade (IIFT), New Delhi

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COURSE INTRODUCTION

Econometrics is a powerful tool for understanding economic relationships and for meaningful research in economics. Economists use econometric methods in a variety of fields including macroeconomics, finance, labour economics, health economics, and many more. Econometric methods are used for business decisions, policy formulation, and prediction.

MECE 101, as the title suggests, is introductory in nature. The basic objective of the course is to equip students with basic theory of econometrics and relevant applications of the methods. The topics covered in the present course include various problems faced in estimation of single equation regression models.

After going through the course, a student is expected to

- develop a way of thinking in econometric terms;
- acquire basic econometric skills in terms of model selection, model estimation, and model diagnostics;
- utilize econometric techniques in analysis and interpretation of real-world problems in the field of economics and business; and
- summarise and interpret results of econometric analysis in written, oral and visual forms; and
- develop skills for future research.

The course comprises four blocks.

Block 1, entitled **Introduction**, comprises three Units. As the title suggests, the block provides an overview and basic foundations of econometrics. Unit 1 discusses the basic concepts, issues and ideas in an econometric study. Unit 2 reviews the statistical concepts required in econometric analysis. Unit 3 provides an overview of matrix algebra, which is essential in econometric theory.

Block 2, entitled **Classical Regression Model**, comprises four units. Unit 4 deals with estimation, interpretation and evaluation of two-variable regression models. Unit 5 helps you in identifying problems in a regression model by looking into the residuals. Unit 6 and Unit 7 extend the regression model to multiple explanatory variables cases. By going through these two Units, you will learn the procedure of estimation and evaluation of multiple regression models.

Block3, entitled **Violations of Basic Assumptions**, analyses the properties of the OLS estimators when certain assumptions are not fulfilled. There are six Units in this Block. Unit 8 brings out the specification issues – the possible problems in a regression model – when the basic assumptions of the classical regression model are violated. Unit 9 deals with the identification, consequences and remedial measures if the error terms are correlated. Unit 10 presents the consequences and remedial measures if there is strong correlation between the explanatory variables. Unit 11 delas with the problem of heteroscedasticity where variance of the stochastic error term is not constant across observations. Unit 12 acquaints you with the

consequences and remedial measures of errors in dependent and independent variables. Unit 13 discusses the issues when the explanatory variable is correlated with the stochastic error term – the endogeneity problem. In this Unit, the student is introduced to the instrumental variable (IV) method and the two-stage least squares (2SLS) method.

Block 4, entitled **Extensions of Regression Models**, is devoted to two issues – qualitative variables, and simultaneity bias. It comprises three Units. Unit 14 deals with the issues related to the introduction of dummy variables as explanatory variables. Unit 15, on the other hand, deals with issues related to the introduction of dummy variable as dependent variable in regression model. In this Unit, students will get to learn about the logit model and the probit model. Unit 16 introduces the students to possible problems if an equation is estimated by the OLS method, when the equation in fact is a part of an equation system. The issues of endogenous explanatory variables and simultaneity bias are highlighted in this Unit.

After going through MECE 101, students may opt for the course MECE 102: Advanced Econometric Methods. As the name suggests, MECE 102 deals with advanced topics such as time series analysis and panel data models.



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UNIT 1 INTRODUCTION TO ECONOMETRICS *

Structure

- 1.0 Objectives
- 1.1 Introduction
- 1.2 Nature of Econometrics
 - 1.2.1 Theoretical Econometrics and Applied Econometrics
 - 1.2.2 Econometric Models
 - 1.2.3 Important Steps in an Econometric Study
- 1.3 Specification of an Econometric Model
 - 2.3.1 Deterministic Relationship
 - 2.3.2 Stochastic Relationship
 - 2.3.3 Inclusion of Error Variable
- 1.4 Data Generation Process
- 1.5 Functional Forms
 - 1.5.1 Classical Regression Model
 - 1.5.2 Intrinsically Linear Cases
- 1.6 Software Packages for Econometric Analysis
- 1.7 Let Us Sum Up
- 1.8 Key Words
- 1.9 Some Useful Books
- 1.10 Answers/Hints to Check Your Progress Exercises

1.0 OBJECTIVES

After going through this Unit, you would be in a position to

- explain why we should study econometrics;
- appreciate the scope of econometrics;
- point out the important steps in an econometric study;
- distinguish between deterministic relationship and stochastic relationship among variables;
- identify reasons for inclusion of error variable in a regression model;